Residual-Rank Entanglement of Conjugate Weil Pairs in the Spectral Admissibility Cascade

Jérôme Beau

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Abstract

We compute the entanglement entropy of a closed conjugate Weil pair $\{c, q-c\}$ along the spectral admissibility cascade of the Cosmochrony programme, where entanglement is the infra-projectable invariant of a non-decomposable joint fibre of the non-injective projection. The bipartite state lives on the residual fibre-level support of the Gram–Schmidt span of Weil fingerprints on the Heisenberg Cayley graph. We show that the normalized admissibility residual of each newly admitted independent direction equals unity: the admitted Weil fingerprints are mutually orthogonal (proved analytically in the companion paper [20]), so the reduced Schmidt spectrum is flat and the reduced entropy is exactly $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$, the logarithm of the residual Gram–Schmidt rank. No Born–Infeld spectral correction survives: the admissibility-weight deficit is identically zero. Because the cascade only adjoins independent directions, $r_{\mathrm{pair}}(n)$ is non-increasing structurally, so the residual conjugate-pair entanglement is maximal at the earliest admissible stage and decreases monotonically as correlated capacity is stabilized into projected structure. The capacity-level statement (non-negativity of the stabilization rate) is established numerically. The entropy computed here is the full residual-rank observable on the complete Gram–Schmidt basis; its relation to the minimal admissible pair sector that carries the Bell–Tsirelson singlet entropy $\log 2$ is established separately [18]. The remaining finite-q task is to measure the contraction law of $r_{\mathrm{pair}}(n)$ and compare its exponent with the pair-capacity exponent δ_{pair}.

Keywords: non-injective projection; Weil representation; Heisenberg group; entanglement entropy; spectral admissibility; conjugate pairs; Schmidt rank; Born–Infeld parity.

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1 Introduction

The Cosmochrony programme reconstructs physical structure from admissibility constraints on non-injective projections on the Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, with the non-injective projection Π as the primitive of physical description [5, 2, 11]. Within this programme, entanglement is not an added postulate. It is the infra-projectable invariant of a non-decomposable joint fibre of Π: two degrees of freedom that share a single resolution cell of Π cannot be assigned separate projected descriptions, and the correlation between them is exactly the part of the joint fibre that Π cannot split. The same non-injective mechanism already yields the violation of Bell factorisation and the Tsirelson bound from the projection-fibre geometry [3], and the phase-coherent correlations of the quantum sector from the failure of factorisation under admissibility [8, 10].

What has been missing is a quantitative location of entanglement *within* the spectral admissibility cascade itself: given the conjugate Weil structure that organises the cascade, how much entanglement does a conjugate pair carry, and how does it evolve as the cascade resolves? This note answers that question for a single closed conjugate pair $\{c, q-c\}$.

The result is structurally simple. The reduced state of the pair, restricted to the residual fibre-level support of the Gram–Schmidt span of Weil fingerprints, has a flat Schmidt spectrum, because the admitted fingerprint directions are mutually orthogonal. Hence the reduced entropy is exactly the logarithm of the residual rank,

$$S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n),$$

with no surviving spectral weighting. This entropy is maximal at the earliest admissible stage and decreases monotonically; the monotonicity is structural, following from the Gram–Schmidt construction alone. The mass hierarchy of the projected description is then the downstream trace of this stabilization, not an independent input [17, 4].

Two subsequent companion papers sharpen this result. The orthogonality of the admitted fingerprints, stated here as the property $w_j = 1$, is proved analytically in [20]: the k=3 Weil fingerprint is a discrete Fourier mode at frequency $A = c_1 b_1 + c_2 b_2 + c_3 b_3 \bmod q$, and distinct modes are orthogonal. The relation between the present full residual-rank entropy and the minimal admissible pair sector is clarified in [18]: the rank-3 admissible projection carries the spin-$\frac{1}{2}$ singlet with entropy $\log 2$, the quantity relevant to Bell–Tsirelson correlations, while $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ is the finer pipeline-level

observable on the full Gram–Schmidt basis. The two coincide only when $r_{\text{pair}}(n) = 2$.

2 Conjugate Weil pairs and the bipartite fibre

Let ρ_c denote the Weil representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ attached to the central character c, acting on $V_c \cong \mathbb{C}^q$. For a nonzero central character c, the conjugate character is $q - c$, and the two representations are related by complex conjugation,

$$V_{q-c} \simeq \overline{V_c},$$

so $\{c, q-c\}$ forms a natural pair closed under the Born–Infeld parity involution. Born–Infeld parity forces the admissible fibre of the pair to be equatorial, fixing the eigenvalue ratio of the projected covariance and binding the two characters into a single admissible object [21, 9].

The bipartite space at BFS depth n is

$$K_c(n) = W_c(n) \otimes W_{q-c}(n),$$

where $W_c(n)$ is the admissible fibre-level subspace generated by ρ_c up to depth n (made precise in Section 3). We treat the pair as a single closed object.

Hypothesis 1 (Closure of the conjugate pair)**.** *The conjugate pair $\{c, q-c\}$ is in a pure joint state on $K_c(n)$, with Schmidt vectors paired diagonally as $e_j \leftrightarrow \overline{e_j}$.*

Hypothesis 1 is not arbitrary: the diagonal pairing is the one selected by the parity involution $V_{q-c} \cong \overline{V_c}$, not an independent assumption about the joint state [21, 7]. It is the precise sense in which a conjugate pair is a closed, infra-projectable object rather than two separately projected fibres. The reduced state of one factor is then well defined, and its entropy is the entanglement of the pair.

3 The residual fibre-level rank

The cascade observable is the projective capacity of a Weil block. For a generic block with central character c, breadth-first search on the Cayley graph $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$ generates, shell by shell, the Weil fingerprint vectors $\rho_c(g)v_0$, and a Gram–Schmidt pass records how many of them are new [19, 6]. Writing δr_n for the number of fingerprint vectors in shell S_n

that are linearly independent of the span built over shells $0, \ldots, n-1$, the per-shell capacity is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}.$$

The conjugate-pair observable is the product $\sigma^{\mathrm{can}}_{\mathrm{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$, built at the level of the raw cumulative spans so that the residual amplitude factor $r(c,q)$ is treated as a normalization artifact rather than physical content [7, 1].

Definition 1 (Residual fibre-level rank)**.** *Let $R(n) = \sum_{m \le n} \delta r_m$ be the cumulative Gram–Schmidt span dimension at depth n, and let R_∞ be its saturated value. The residual fibre-level rank is*

$$r_{\mathrm{pair}}(n) = R_\infty - R(n),$$

the number of independent admissible directions not yet activated at depth n.

At complete locking $R(n) = R_\infty$, so $r_{\mathrm{pair}}(n) = 0$: this is the trivial separable endpoint. Throughout we read the entanglement entropy with the convention

$$S_{\mathrm{ent}}(n) = \begin{cases} \log r_{\mathrm{pair}}(n), & r_{\mathrm{pair}}(n) \ge 1, \\ 0, & \text{at complete locking}, \end{cases}$$

equivalently using the effective rank $r^{\mathrm{eff}}_{\mathrm{pair}}(n) = \max(r_{\mathrm{pair}}(n), 1)$, so that the separable endpoint carries zero entanglement.

Equivalently, tagging each admitted direction j by its admission depth m_j, the residual support is $\mathcal{R}_n = \{\, j : m_j > n \,\}$ and $r_{\mathrm{pair}}(n) = |\mathcal{R}_n|$. By Hypothesis 1, $r_{\mathrm{pair}}(n)$ is the Schmidt rank of the conjugate pair at depth n (the pair rank is the smaller of the two conjugate factors, which coincide by parity).

4 Reduced entropy: saturation of the rank bound

The admissibility-weight ansatz becomes trivial on the actual Gram–Schmidt data. For each newly admitted Weil fingerprint direction, the normalized residual at the moment of admission satisfies

$$w_j = \frac{\|w_j^{\mathrm{GS}}\|}{\|v_j\|} = 1$$

in the tested finite-q runs. Every admitted direction is thus exactly orthogonal to the previously generated span. This is not imposed by Gram–Schmidt, which would also admit a direction of partial overlap; it is a measured property of the Weil fingerprints, whose distinct images are mutually orthogonal until they become linearly dependent. The admissibility process is therefore binary at the level of independent directions: a fingerprint is orthogonal to the current span and admitted with $w_j = 1$, or already contained in it and rejected.

Proposition 1 (Flat residual Schmidt spectrum)**.** *If the admitted Weil fingerprint directions are mutually orthogonal, the normalized Schmidt weights on the residual support are flat,*

$$p_j(n) = \frac{1}{r_{\mathrm{pair}}(n)}, \qquad j \in \mathcal{R}_n,$$

and the reduced entropy of the closed conjugate pair is exactly

$$S_{\mathrm{ent}}(n) = -\sum_{j \in \mathcal{R}_n} \frac{1}{r_{\mathrm{pair}}(n)} \log \frac{1}{r_{\mathrm{pair}}(n)} = \log r_{\mathrm{pair}}(n).$$

Equivalently the admissibility-weight deficit vanishes,

$$\varepsilon_{\mathrm{adm}}(n) := \log r_{\mathrm{pair}}(n) - S_{\mathrm{ent}}(n) = 0.$$

This equality is not an additional maximal-entanglement postulate. It follows from the orthogonality of the admitted Weil fingerprint directions in the Gram–Schmidt cascade. The identity $w_j = 1$ was first verified across $q \in \{29, 61, 101, 151\}$ and multiple generic blocks, and is now proved analytically in the companion paper [20]: the k=3 Weil fingerprint is a discrete Fourier mode at frequency $A = c_1 b_1 + c_2 b_2 + c_3 b_3 \bmod q$, and a newly admitted frequency is orthogonal to the entire prior span, so $w_j = 1$ exactly.

5 Monotonicity of the entanglement profile

Since the Gram–Schmidt cascade only adjoins independent directions, $R(n)$ is non-decreasing. Therefore $r_{\mathrm{pair}}(n) = R_\infty - R(n)$ is non-increasing.

Proposition 2 (Structural monotonicity)**.** *The entanglement entropy $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ is monotone non-increasing in n. The residual conjugate-pair entanglement is maximal at the earliest admissible stage and decreases as independent fingerprint directions are stabilized into the projected span.*

This rank monotonicity is structural: it follows from the Gram–Schmidt construction itself, with no numerical input. A separate statement concerns the capacity-level stabilization rate

$$\Delta I(n) = \sigma^{\mathrm{can}}_{\mathrm{pair}}(n) - \sigma^{\mathrm{can}}_{\mathrm{pair}}(n+1),$$

which measures the rate at which residual correlated capacity is converted into projected stable structure. The non-negativity of $\Delta I(n)$ — one-way activation at the level of the capacity — is established numerically in the projective temporal ordering analysis, with no physically meaningful regressions observed; a general analytic proof remains open [12]. The monotonicity of the entanglement profile does not rely on this capacity-level result: it is already guaranteed by Proposition 2.

6 Physical interpretation

The conjugate-pair entanglement is maximal at the earliest admissible stage, where the residual rank is largest, and it decreases monotonically as the cascade resolves. There is no intermediate maximum: the trajectory is rank-dominated and, under the orthogonality of the admitted fingerprints, rank-determined. The decrease is not a loss of correlation into an environment; it is the conversion of residual correlated capacity into projected stable structure. The projected mass hierarchy is the downstream trace of this stabilization, recorded by the capacity exponent rather than by an independent dynamical input [17, 14, 4].

The interpretation is consistent with the relational reading of time in the programme. A fully infra-projectable invariant has no position in the projected temporal ordering, which is itself the cumulative projected capacity [12]: it is a single unsplit object below the resolution of Π, not a pair of separately ordered events. Correlations of this kind therefore carry no signal and admit no influence between separated projected descriptions; locality at the projected level is preserved, in the same way that the non-injective mechanism reproduces the Tsirelson bound without superluminal signalling [3]. Entanglement is, in this sense, the not-yet-time-ordered sector of the cascade.

7 Numerical deliverable

The finite-q calculation reduces to the residual fibre-level rank trajectory:

$$r_{\mathrm{pair}}(n), \qquad S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n), \qquad \Delta I(n) = \sigma^{\mathrm{can}}_{\mathrm{pair}}(n) - \sigma^{\mathrm{can}}_{\mathrm{pair}}(n+1).$$

For $q \in \{61, 151, 211, 307\}$, the rank trajectory is reconstructed from the cumulative capacity profiles already produced by the systematic pair campaign [16], via $r_{\mathrm{pair}}(n) = R_\infty - \sum_{m \le n} \sigma_c(m)\,|S_m|$, and $\Delta I(n)$ follows directly from the stored $\sigma^{\mathrm{can}}_{\mathrm{pair}}$ profile.

The task is therefore not to rescue an entropy correction — there is none — but to measure the contraction law of $r_{\mathrm{pair}}(n)$, its scaling exponent, and its relation to the pair-capacity exponent δ_{pair}. If the contraction exponent agrees with δ_{pair} within the expected finite-q window, the entanglement trajectory and the capacity law share the same spectral engine. If it differs, the difference marks a genuine separation between residual rank contraction and shell-wise capacity decay.

8 Remark on possible non-rank corrections

The equality $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ uses the native Gram–Schmidt admissibility weights on Weil fingerprint directions, which are uniform. A non-trivial entropy deficit would require an additional weighting structure on the residual support, for example from the projected $\mathfrak{su}(2)$ coordinates of the admissible fibre or from an explicitly diagonalized Laplacian sector [13, 15]. Such a correction would be a distinct modelling choice, defined on a different observable, and is not part of the present fibre-level entanglement calculation. That different observable is the minimal admissible pair sector of [18]: the rank-3 admissible projection, spectrally atomic, whose two-dimensional parity pair carries the spin-$\frac{1}{2}$ singlet with entropy $\log 2$. It is the sector relevant to the Bell–Tsirelson correlator, and is distinct from the full residual-rank observable computed here.

9 Conclusion

The entanglement of a closed conjugate Weil pair, computed honestly on the spectral admissibility cascade, returns to a pure O-series observable: the logarithm of the residual fibre-level rank. The chain is

$$\begin{aligned} &\text{closed conjugate pair} + \text{orthogonal admitted fingerprints} \\ &\quad\Longrightarrow\ \text{flat Schmidt spectrum} \\ &\quad\Longrightarrow\ S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n), \end{aligned}$$

with monotone decrease guaranteed structurally by the Gram–Schmidt construction. No free parameter and no external weight enters.

The epistemic status is stratified as follows. The reduction $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ is derived, conditional on the orthogonality of the admitted fingerprints, which is numerically established for $q \in \{29, 61, 101, 151\}$ and proved analytically in the companion paper [20]. The monotonicity of S_{ent} is structural. The capacity-level one-way activation governing ΔI is numerically established in the projective temporal ordering analysis, with a general analytic proof open. The purity of the conjugate pair (Hypothesis 1) rests on the Born–Infeld parity involution and is the standing structural assumption of the construction. The entropy computed here is the full residual-rank observable; its separation from the minimal admissible pair sector ($S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$, the Bell–Tsirelson quantity) is treated in [18].

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